

Supplementary Material

Supplementary Table S1. Corrected quantitative claim-to-artifact map

Manuscript claim / quantity	Artifact	Field / row used
Corrected development Geneformer AUROC and interval	corrected_v1_descriptive.json	modes.shared.development_geneformer
Development pseudobulk full-space AUROC, CI, permutation p	l2_dev_sle_vs_healthy.csv	row pseudobulk; same performance fields
Development age AUROC, CI, permutation p	l2_dev_sle_vs_healthy.csv	row metadata_only_age; same performance fields
Corrected Geneformer C selections and final C	corrected_v1_shared_fit_report.json	development_selected_c_per_outer_fold; final_c_median
Historical Geneformer frozen C, retained	FREEZE.json	hyperparameters.geneformer.C_per_outer_fold; C_frozen
Frozen pseudobulk C selections	FREEZE.json	hyperparameters.pseudobulk.C_per_outer_fold = [0.01, 0.1, 0.1, 0.01, 0.01]; C_frozen = 0.01
Frozen age C selections	FREEZE.json	hyperparameters.metadata_only_age.C_per_outer_fold = [0.001, 0.001, 0.001, 0.001, 0.001]; C_frozen = 0.001
External AUROC, AP, Brier, calibration	corrected_v1_shared_analysis.json	metrics.<arm>; also corrected_v1_descriptive.json for AUROC CIs
Corrected paired AUROC differences, intervals, DeLong/Holm	corrected_v1_shared_analysis.json	comparisons.geneformer_vs_age; geneformer_vs_pseudobulk
Historical decision terminology	l2_sealed_results.json	REJECTED in original release; Not met in revised manuscript
Corrected Geneformer cohort-membership probe	corrected_v1_shared_cohort_probe.json	cohort_membership_auroc; ci95
Development/external age and development sex/ancestry summaries	l2_cohort_confounders.csv	cohort/variable/category rows
Ten reseeded nested-CV AUROCs and selected C values	l2_dev_repeated_cv.csv	all 10 rows per arm; manuscript reports mean and SD across repeats
Corrected external age strata	corrected_v1_descriptive.json	modes.shared.external.<arm>.Children/Adult
V1 extraction and donor/cell accounting	shared_features/	two *_run_summary.json and ingestion_validation.json files
Model-native Geneformer sensitivity	corrected_v1_native_analysis.json	metrics.geneformer; comparisons
Native minus shared input paired effect	corrected_v1_gene_input_sensitivity.json	native_minus_shared
Gene availability diagnostic	historical_gene_availability.json	shared_genes; development_zero_but_external_expressed_genes

Manuscript claim / quantity	Artifact	Field / row used
External sample crosswalk and age source sensitivity	external_age_source_sensitivity.json	source_records; age_conflict; shared_geneformer_minus_sensitivity_age
Original-logit Geneformer calibration	corrected_v1_shared_analysis.json	metrics.geneformer.calibration.score_convention; intercept; slope; CI

Corrected artifact names in Table S1 are relative to results/correction_2026-09; historical l2_* files are relative to results/, and FREEZE.json is at repository root. The original release and preregistration remain visible. Its historical decision token REJECTED is rendered as “Not met” for the prespecified Geneformer-superiority criterion. This correction used the previously examined external cohort.

Supplementary Table S2. Age-stratified external-cohort performance

Arm	Source Children (n = 44)	Source Adult (n = 12)
Geneformer	0.7052 (0.5452–0.8533)	0.8857 (0.6250–1.0000)
Pseudobulk	0.8926 (0.7812–0.9746)	0.9143 (0.6944–1.0000)
Metadata (age)	0.7658 (0.5534–0.9410)	0.4286 (0.0833–0.7857)

Values are AUROC (95% percentile donor-bootstrap CI), based on 5000 draws of fixed predictions. The source Adult stratum had 7 SLE and 5 healthy samples; 10 of 5000 bootstrap draws were single-class and discarded. The source Children group includes some donors aged 18 or 19. Prespecified stratum-specific permutation and co-primary tests were not completed and are reported as protocol deviations.

Supplementary provenance and sensitivity

Shared-gene main analysis: 261 development donors and 1,263,676 cells; 56 external samples and 363,083 cells; 256-dimensional finite donor vectors. The four archived run summaries record the pinned Geneformer revision, checkpoint-file and embedding SHA-256 hashes, V1 settings, cell accounting, and complete runtime package inventories. The eight-cell fixture and 30,165-gene intersection were checked. Three V1 dictionary hashes were printed during successful Kaggle setup but not exported in the four original share ZIPs; they are user-reported setup evidence, not independently archived run-summary fields. Revised notebooks export a machine-readable setup and checkpoint provenance record for future runs. The original ZIPs were retained outside the repository.

Model-native input was a labeled exploratory sensitivity, not a replacement primary analysis. Development Geneformer AUROC was 0.9754 (95% CI 0.9573–0.9898); external AUROC was 0.7484 (0.6122–0.8699). Native minus shared external AUROC was +0.0141 (paired 95% CI –0.0238 to +0.0597; two-sided paired DeLong $p = 0.4601$). Native Geneformer minus age was +0.1703 (Holm $p = 0.1595$); native Geneformer minus pseudobulk was –0.1500 (Holm $p = 0.0181$). Neither native comparison met a positive Geneformer-superiority rule.

Calibration is diagnostic and did not alter predictions. External Brier scores were 0.4713 for corrected shared-gene Geneformer, 0.1489 for pseudobulk, 0.2156 for age, and 0.2128 for a constant probability fixed at development SLE prevalence (162/261). The saved original Geneformer decision scores yield recalibration intercept 2.091 (95% Wald CI 0.792–3.390) and slope 0.149 (0.027–0.271). Reconstructed logits after clipping eight Geneformer probabilities to $[10^{-6}, 1-10^{-6}]$ instead yield intercept 2.374 and slope 0.185. Pseudobulk and age slopes are 0.545 and –5.414 from clipped probability logits; one

pseudobulk probability was clipped and age is unstable. These are exploratory diagnostics on the previously examined cohort.

External metadata reconciliation. The dated GEO series SOFT record contains 56 accessions. Comparing its source cohort naming scheme with repository metadata identifies aHD2 and aSLE8 as the two source-described labels absent from the analyzed deposit; their absence is unexplained. GEO sample GSM4029942 (aHD3, JB19003; 7,206 analyzed cells) records age 50. An independent review reported age 43 in the publication supplement, which was unavailable here for independent hash verification. Primary age scoring retains GEO 50 by the prespecified source-priority rule. Substituting 43 with the same fitted age model changes overall age AUROC from 0.5781 to 0.5797 and Adult age AUROC from 0.4286 to 0.4571. Shared Geneformer minus age becomes +0.1547 (paired 95% CI -0.0862 to +0.4022; paired DeLong $p = 0.2082$); the superiority rule remains unmet. The 58-row source-to-GSM crosswalk and GEO source hashes are versioned in the repository. Supplementary Table 4 of the source study reports individual-level cluster and subcluster counts, but their correspondence to exact barcode-level labels in these raw matrices was not established.

Supplementary reporting and bias assessment

TRIPOD+AI reporting items were mapped to the revised title/abstract, source and eligibility, donor-level outcome and predictors, missing age handling, model fitting and validation, participant accounting, discrimination, calibration, open code, and limitations. The repository audit records each location and remaining gap. A PROBAST+AI-informed appraisal flags high risk of bias or major applicability concern for clinical prediction because the external sample is small and demographically different, correction followed examination of the external outcomes, pretraining overlap is unknown, and clinical-use thresholds and utility were not evaluated. This is an author-conducted assessment, not an independent adjudication.